

# Rahul Tamanam

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## SKILLS

**Data Engineering:** Python, R, SQL, ETL/ELT, Snowflake, Data Pipelines, Data Modeling, Cost Optimization, PySpark, Hadoop(HDFS), Hive, Impala

**Databases and Tools:** MySQL, SQLite3, PostgreSQL, MongoDB, Git, Docker, FastAPI

**Cloud Platforms:** AWS (S3, Redshift, Glue, Lambda, Athena), GCP (BigQuery, Dataflow, Cloud Storage)

**Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Power BI, Tableau, Excel

**AI/ML & LLM Tools :** Groq, LangChain, Pinecone, Neo4j, Ollama, Anthropic Claude API, Hugging Face (PEFT, LoRA fine-tuning), vLLM, Axolotl, Weights & Biases, Scikit-learn, XGBoost, spaCy

## EXPERIENCE

### Student Assistant, University Housing Operations

April 2025 – Present

*The University of Texas at Dallas*

*Richardson, TX*

- Managed 100+ weekly resident requests by logging maintenance tickets, resolving housing inquiries, and coordinating check-in/check-out operations, ensuring accurate records and timely issue resolution within the housing systems
- Developed Excel-based operational data systems leveraging formula-driven automation, and dynamic data organization to track room assignments and maintenance schedules for 800+ residents, improving reporting efficiency by 25%

### AI/ML Intern

January 2024 – May 2024

*HMI Engineering Services*

*Visakhapatnam, AP*

- Built a predictive maintenance prototype by cleaning and aggregating time-series sensor data, engineering failure related features, and training classification models using Python, pandas, scikit-learn, and SQL
- Analyzed model predictions to identify high risk equipment and worked with engineers to adjust maintenance thresholds, increasing average equipment uptime by 10%
- Developed failure trend visualizations and sensor drift analyses using Matplotlib and Seaborn to communicate degradation patterns across 6 months of equipment data
- Contributed to an estimated 8% reduction in annual maintenance spend by enabling earlier interventions and fewer emergency repairs

## PROJECTS

### CometBot | Python, FastAPI, Neo4j, Pinecone, Typescript

January 2026 - May 2026

- Built CometBot, an AI advising platform for UTD JSOM graduate students, enabling degree planning, career guidance, and skills gap analysis using React, FastAPI, Neo4j, and Pinecone with Groq-powered LLM reasoning
- Designed a hybrid deterministic + RAG architecture where a Neo4j graph enforces academic constraints and degree progression, while Pinecone retrieval grounds LLM outputs, improving reliability of course recommendations
- Replaced client-side PDF parsing with a server-side onboarding pipeline, automatically extracting structured academic data and mirroring production-grade advising workflows

### Velox | React, Node/Express, SQLite, Tailwind CSS, Recharts

May 2026

- Built a macro-aware portfolio engine using FRED (GDP, Fed Funds Rate) to classify market regimes and rebalance an ETF universe monthly, benchmarked against SPY with regime-shaded performance charts
- Ran 1,000-path Monte Carlo simulations via Geometric Brownian Motion, computing p10–p90 fan charts, max drawdown, CAGR, and goal-attainment probability from 5 years of per-ticker price history
- Developed a full-stack investment platform (React, Express, SQLite) with a composite Health Score, live Yahoo Finance integration, LLM-powered assistant, and a portfolio shock simulator

### Churn Intelligence System | Python, SQL, XGBoost, FastAPI, React, LLMs

February 2026 – March 2026

- Built an end-to-end churn pipeline on 30M+ behavioral events, engineering 40+ features in DuckDB (sessionization, 30/60/90-day decay, cohort segmentation) and training an XGBoost model on 860k users with 5-fold CV achieving production-grade accuracy
- Implemented a T-learner uplift model to isolate persuadable users via incremental treatment effects, layered PELT change-point detection for behavioral regime shifts, and deployed SHAP + LLM-generated retention recommendations

## EDUCATION

### The University of Texas at Dallas

May 2026

*Master of Science in Business Analytics and Artificial Intelligence*

### Gayatri Vidya Parishad College of Engineering

May 2024

*Bachelor of Technology in Information Technology*